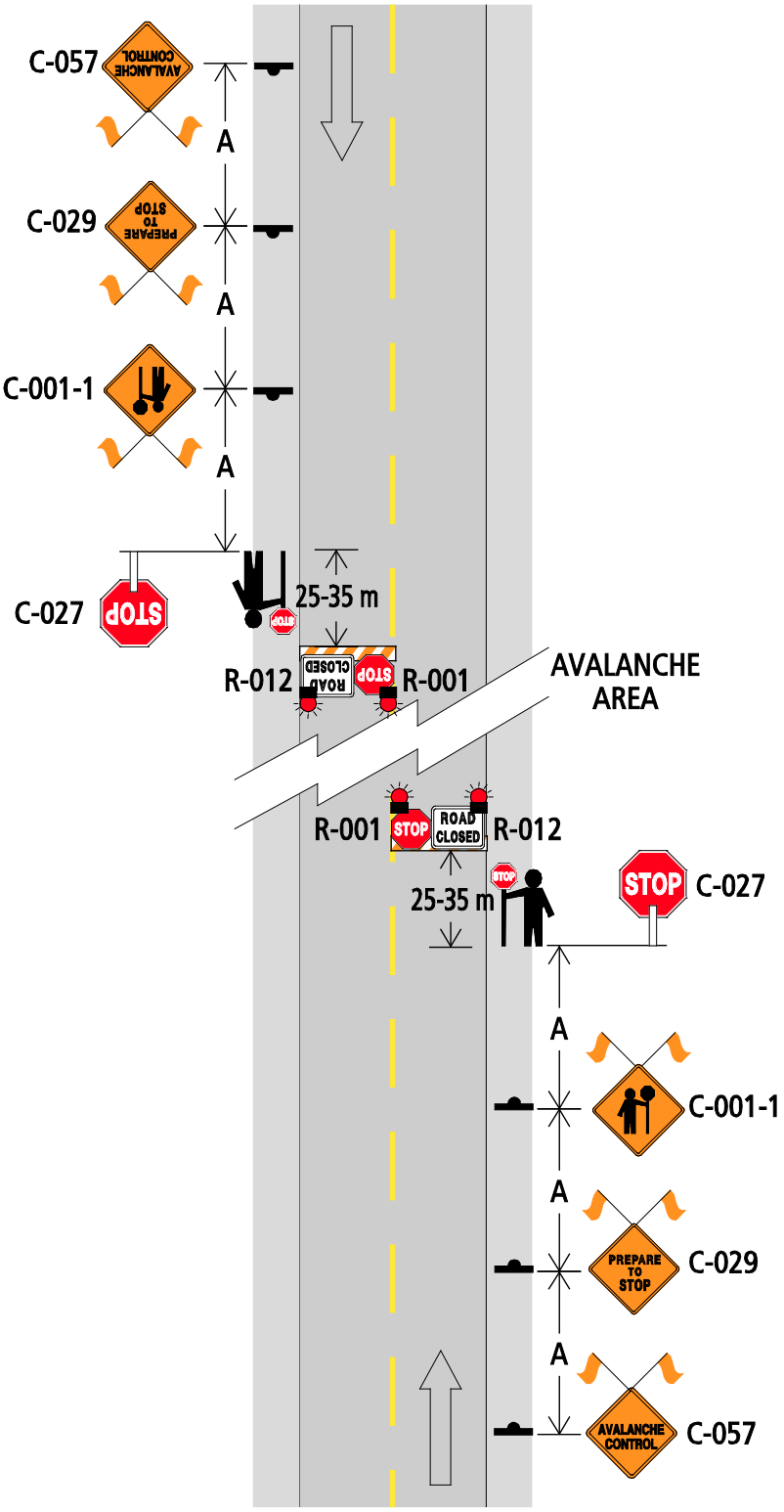
Figure 16.4: Road Closure for Avalanche Control – Barricades and TCPs



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PART B – TRAFFIC CONTROL

Section 17: Traffic Control Layouts – Utility Work

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– Short and Long Duration

17.3 Traffic Signal Relamping/Cleaning 17-6
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PART B – TRAFFIC CONTROL

Section 17: Traffic Control Layouts – Utility Work

LEGEND			
	Flexible Drum		Paint Truck
	Tubular Marker		Escort Truck
	Cone		Chaser Vehicle
	Sign		Vehicle Mounted Rear Crash Attenuator
	Traffic Control Person		
	Work Activity Area		360° Flashing Light
	Work Truck		Portable Traffic Signal
	Shadow Vehicle		Barricade and Fencing
	Shadow Vehicle #1		Flashing Arrow Board (FAB)
	Shadow Vehicle #2		Flashing Arrow Board (FAB) in caution mode
	Buffer Vehicle		Dynamic Message Sign (DMS)

TABLE A – TAPER LENGTHS

Taper Types (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Merge Taper Length	L_M	35	55	160	190	210	230	250	280
Lane Shift Taper Length	L_L	30	50	80	100	110	120	130	140
Downstream Taper Length	L_D	30	30	30	30	30	30	30	30
TCP, Signal, and Shoulder Taper Length (min. 5 devices)	L_S	5	8	15	15	15	15	15	15
Minimum Tangent Length between Tapers	L_T	30	60	160	190	210	230	250	280
Run-In Length on Centreline	L_R	40	50	60	60	70	80	90	100

TABLE B – DEVICE SPACING LENGTHS

Device Spacing (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Construction Sign Spacing	A	40	60	80	100	150	150	200	200
Buffer Space	B	30	40	60	80	110	140	170	200
Roll-Ahead Buffer Distance	R	30	30	40	40	40	50	50	50
Channelizing Device Spacing for Tapers	C	10	10	15	15	15	15	15	15
Channelizing Device Spacing on Curves and Tangents	D	10	10	30	30	40	40	40	50

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17.1 General Information - Utility Work

1. The traffic control required for utility work on roadways is typically no different from that required for road construction or maintenance on roadways.
2. The layouts in Section 17 are designed for utility work in a low-speed, low-volume setting.
3. When these layouts are not sufficient for the identified work—that is, if the work is being done in a higher speed or higher volume environment, or if it is related to overhead power, phone, or fibre optic lines—the other Traffic Control Layouts in the Manual (i.e., those in Sections 7 to 18) shall be considered, and the appropriate ones applied.

17.2 Utility Work on Centreline – Urban Area – Short and Long Duration

Purpose:

Various utilities may be situated within the roadway, including manholes and catch basins.

This layout is intended for utility works that can take place in a single shift on low-speed urban roadways.

Standard:

- For short-duration work, a Crew Working Ahead C-004 sign is required.
- For long-duration work, a Construction Ahead C-018-1A sign is required.
- Where the posted speed limit is ≥ 70 km/h, a flashing arrow board (FAB) is required.
- For two-lane, two-way roadways, at least 3 metres of roadway shall remain available on each side of the work area for vehicles to pass on either side. If this cannot be achieved on each side, single lane alternating traffic is required.
- For multilane roadways, lane closures are required.
- Night work may be required because of the site conditions, in which case traffic control signage and devices shall be increased to include:
  - tubular markers instead of cones
  - flashing yellow lights on signs instead of flags

Guidance:

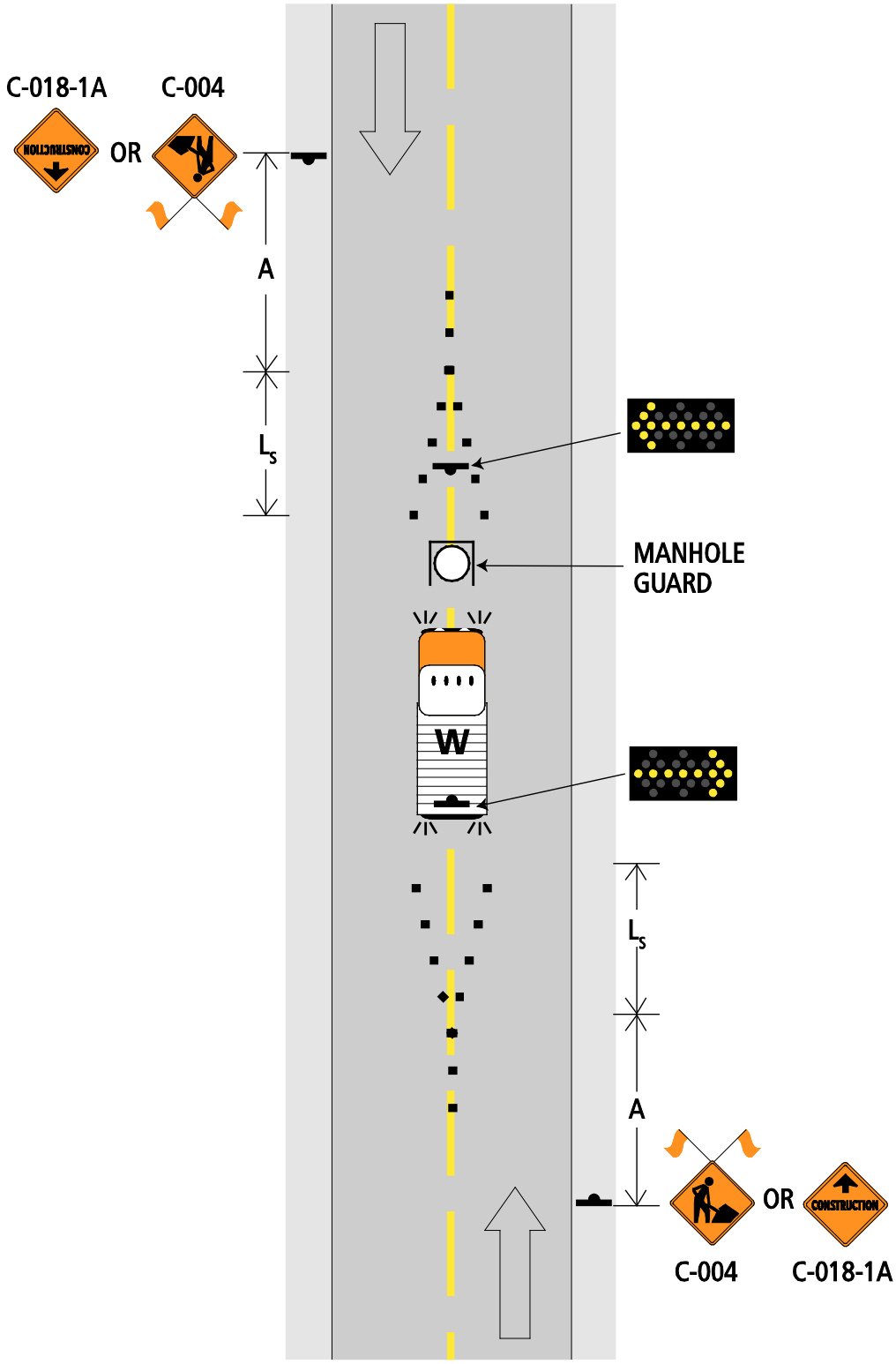
- A Crew Working – Maximum Speed C-002-2 sign and a Thank You Resume Speed C-086-1 sign should be used only where conditions warrant.

Options:

- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 17.2: Utility Work on Centreline – Urban Area – Short and Long Duration



17.3 Traffic Signal Relamping/Cleaning – Short Duration

Purpose:

Electrical utilities within a signalized intersection periodically require the installation of a new signal head or lamps, or cleaning.

This layout is for short-duration work on low-speed, low-volume roadways.

For work on high-volume roadways, which require more complex layouts, and for night work (long-duration work), see *Section 11: Traffic Control Layouts – Intersections.*

Standard:

- A work vehicle equipped with 4-way flashers shall be stopped under a signal head where lamps are to be replaced or cleaned. It requires a flashing arrow board (FAB) that directs traffic into the right or left lane.
- A Crew Working Ahead C-004 sign shall be used on all approaches to the intersection.
- If Traffic Control Persons are used, they shall never provide direction that conflicts with that provided by a traffic signal. Traffic shall be directed by a police officer or the signal shall be turned off or covered.

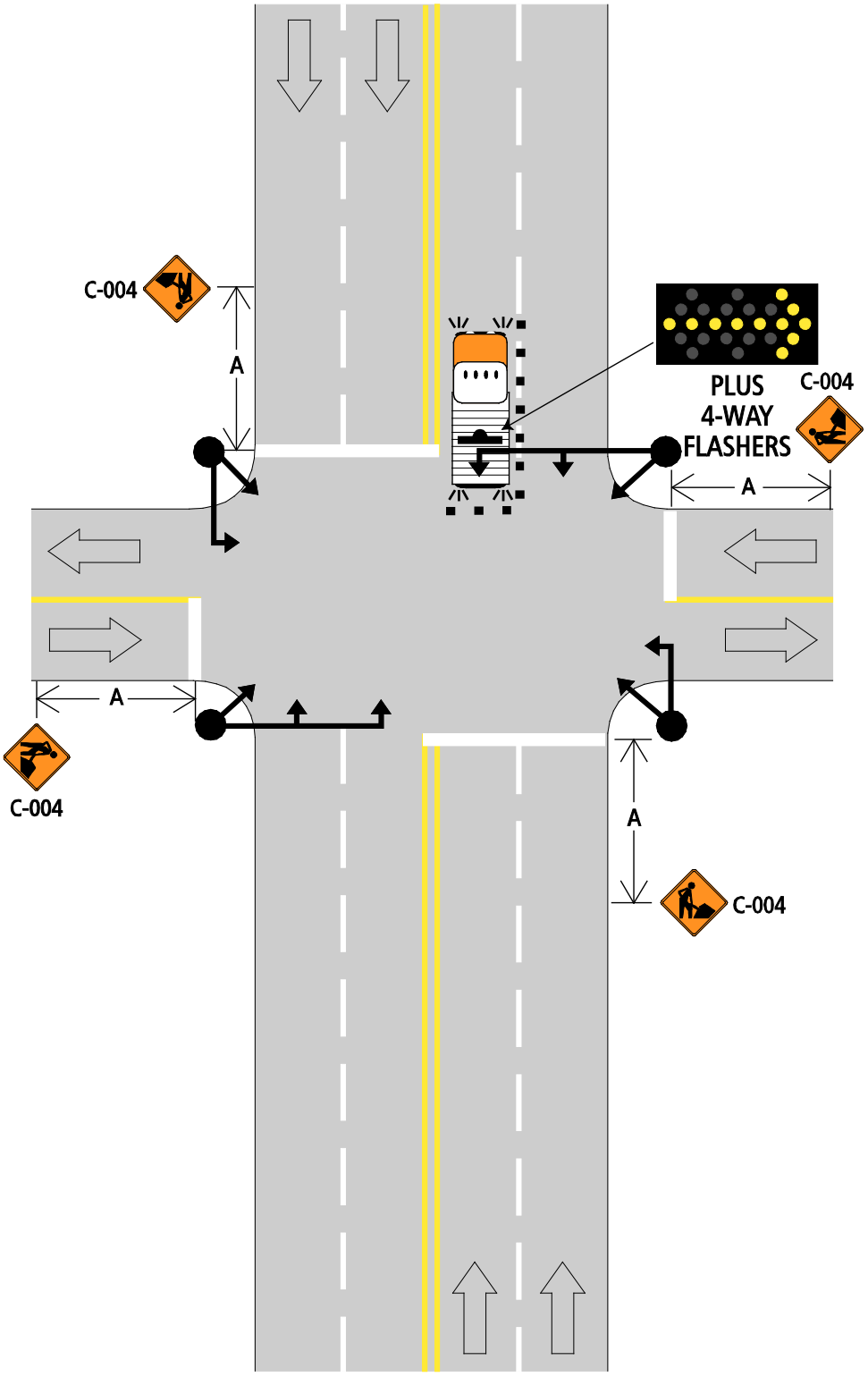
Guidance:

- None at this time.

Options:

- None at this time.

Figure 17.3: Traffic Signal Relamping/Cleaning – Short Duration



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PART B – TRAFFIC CONTROL

Section 18: Traffic Control Layouts – Bicycle Lanes

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PART B – TRAFFIC CONTROL

Section 18: Traffic Control Layouts – Bicycle Lanes

LEGEND			
	Flexible Drum		Paint Truck
	Tubular Marker		Escort Truck
	Cone		Chaser Vehicle
	Sign		Vehicle Mounted Rear Crash Attenuator
	Traffic Control Person		
	Work Activity Area		360° Flashing Light
	Work Truck		Portable Traffic Signal
	Shadow Vehicle		Barricade and Fencing
	Shadow Vehicle #1		Flashing Arrow Board (FAB)
	Shadow Vehicle #2		Flashing Arrow Board (FAB) in caution mode
	Buffer Vehicle		Dynamic Message Sign (DMS)

TABLE A – TAPER LENGTHS

Taper Types (m)		Regular Posted Speed Limit (km/h)							
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Lane Shift Taper Length	L_L	30	50	80	100	110	120	130	140
Downstream Taper Length	L_D	30	30	30	30	30	30	30	30
TCP, Signal, and Shoulder Taper Length (min. 5 devices)	L_S	5	8	15	15	15	15	15	15
Minimum Tangent Length between Tapers	L_T	30	60	160	190	210	230	250	280
Run-In Length on Centreline	L_R	40	50	60	60	70	80	90	100

TABLE B – DEVICE SPACING LENGTHS

Device Spacing (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Construction Sign Spacing	A	40	60	80	100	150	150	200	200
Buffer Space	B	30	40	60	80	110	140	170	200
Roll-Ahead Buffer Distance	R	30	30	40	40	40	50	50	50
Channelizing Device Spacing for Tapers	C	10	10	15	15	15	15	15	15
Channelizing Device Spacing on Curves and Tangents	D	10	10	30	30	40	40	40	50

18.1 General Information on Accommodating Cyclists

Under the Motor Vehicle Act, cyclists have the same rights and duties as operators of motor vehicles. Therefore, the signage and traffic control measures used for drivers also applies to cyclists.

The potential for cyclists to respond differently than drivers under certain conditions should be anticipated and considered. For example, road surface conditions can impact cyclists more so than motor vehicles.

The following factors shall be considered when establishing or inspecting a work zone that may be traversed by cyclists:

- Cyclists are vulnerable road users who have little protection from falls and collisions.
- Cyclists ride on two narrow tires.
- The loss of traction or deflection of the front bicycle wheel can cause a fall.
- Loose gravel, uneven surfaces, milled pavement, and tack coats can create problems for cyclists.
- In the dark, the limitations of bicycle lights make temporary road works difficult for cyclists to see.
- Road works on urban roadways may affect a large variety of cyclists with varying abilities, ranging from children to commuters.

Specific ways to accommodate cyclists during road works are addressed below.

18.1.1 Roadway Surface Conditions

Since cyclists operate on two narrow tires, the quality of the road surface is more important than it is for cars and trucks.

Wherever possible, a smooth, hard surface (such as asphalt) should be maintained for cyclists. Compacted gravel may be acceptable for temporary conditions.

Loose or uneven surfaces should be avoided, and signs indicating bumps or changes in the roadway surface should be used to notify cyclists.

18.1.2 Roads with Bike Lanes or Paved Shoulders

Where cyclists approach the work zone in a bike lane or on a paved shoulder, it is preferable to maintain those facilities within the work zone.

At temporary facilities, a minimum 1-metre width may be used as the space for riding cyclists, with a 0.3-metre clearance from vertical obstacles like cones or barriers.

If there is insufficient space for bicycles and motor vehicles to operate side-by-side, a temporary shared lane may be created. Cyclists and motorists can share a lane over a short distance in low-speed environments (≤ 60 km/h).

Notification should be provided that the roadway is a shared roadway and that the lane is too narrow for side-by-side operation. Cyclists and motorists should operate single file (e.g. Cyclist Right-of-Way Take the Lane C-184 sign).



C-184
Cyclist Right-of-Way
Take the Lane

18.1.3 Roads with Shared Lanes

A shared lane should be used where the width required for a separate bike lane or paved shoulder cannot be maintained.

In low-speed environments (≤ 60 km/h), a shared lane width less than 4.3 metres may be used. Where lane widths are less than 4.0 metres, a shared lane may not be feasible because drivers may have to enter the oncoming lane in order to pass cyclists.

Where the shared lane width is less than 4.0 metres, consider prohibiting motor vehicles from passing cyclists (i.e., single file vehicle/bicycle operation) or detouring cyclists.

Where speeds exceed 60 km/h, a shared lane width of 4.3 metres should be maintained. If this is not possible, it may be necessary to detour cyclists.

18.1.4 Cyclist Detours

Accommodating cyclists within the work zone is preferable to detouring them. Bicycles require less space than cars and can often be accommodated even when vehicles must be detoured. Cyclists can share a lane over a short distance in low-speed environments (≤ 60 km/h).

Requiring cyclists to dismount should be avoided but may be preferable to a lengthy detour. The "cyclists dismount and walk" measure should be considered only if the reason for dismounting is immediately apparent to cyclists. Otherwise, many cyclists may ignore this instruction. Generally, cyclists should not be expected to dismount and walk for more than 50 metres.

Detours should be established where sufficient width or roadway surface for cyclists cannot be maintained through the work zone.

Cyclist detour signs (B-C-004 series) should be used to mark cyclist-specific detour routes and guide cyclists along the route.

Detours in which cyclists have to make left turns or cross arterial roadways without a signal or push-button crosswalks are discouraged.



B-C-004-1A
Bicycle Detour Ahead

Detours are usually not a realistic option for cyclists on rural highways because there may be few alternatives, and detours using alternative routes may cover substantial distance.

18.1.5 Signs and Other Provisions for Cyclists

Signs and other provisions to accommodate cyclists will vary with the nature of the road work, but may include:

- use of dynamic message signs (DMS) or customized signs to forewarn cyclists of construction activity
- use of a Bicycles/Pedestrians Slow B-C-020 series sign within the advance signing array to advise drivers and cyclists to slow down
- use of a Share the Road W-132-1 series sign to advise drivers that cyclists will be using the travelled roadway
- use of appropriate pavement surface condition signs (e.g., Grooved Pavement C-011 sign)
- regular sweeping of the shoulder near the active work site to reduce debris from construction activity
- providing cyclists with a ride through the work site when pilot vehicles are present
- consultation with local cycling organizations
- detouring cyclists (may not be feasible in rural areas)

The traffic control layouts are considered the minimum standard. The associated text description of each figure highlights the key standards as well as guidance and options that can be considered by the user. The following information is typical on most layouts in this section:

Standard:

- For short-duration work, a Crew Working Ahead C-004 sign is required.
- For long-duration work, a Construction Ahead C-018-1A sign is required.
- A Bike Lane Closed B-C-002 sign is required in advance of the bicycle lane closure.

Guidance:

- None at this time.

Options:

- A dynamic message sign (DMS) may be used for additional messaging if space allows.
- In low speed (≤ 60 km/h) urban areas (within municipal boundaries), customization of traffic control layouts, including closer device spacing and shorter taper lengths, may be necessary to maintain access and mobility. In these cases, document why adjustments are being made.

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18.2 Bicycle Lane Shift

Purpose:

This layout shows the typical setup of a bicycle lane shift where the road width allows bicycle traffic to be maintained through the work area by shifting all vehicle and bicycle traffic and still maintaining a separated lane for bicycles.

This layout does not apply for shoulder closures.

Standard:

- A Lane Shift C-117-L/R sign shall be used in advance of the shifting lanes.
- A Lane Closure Arrow C-053 sign shall be used where the bicycle and vehicle traffic is being shifted.

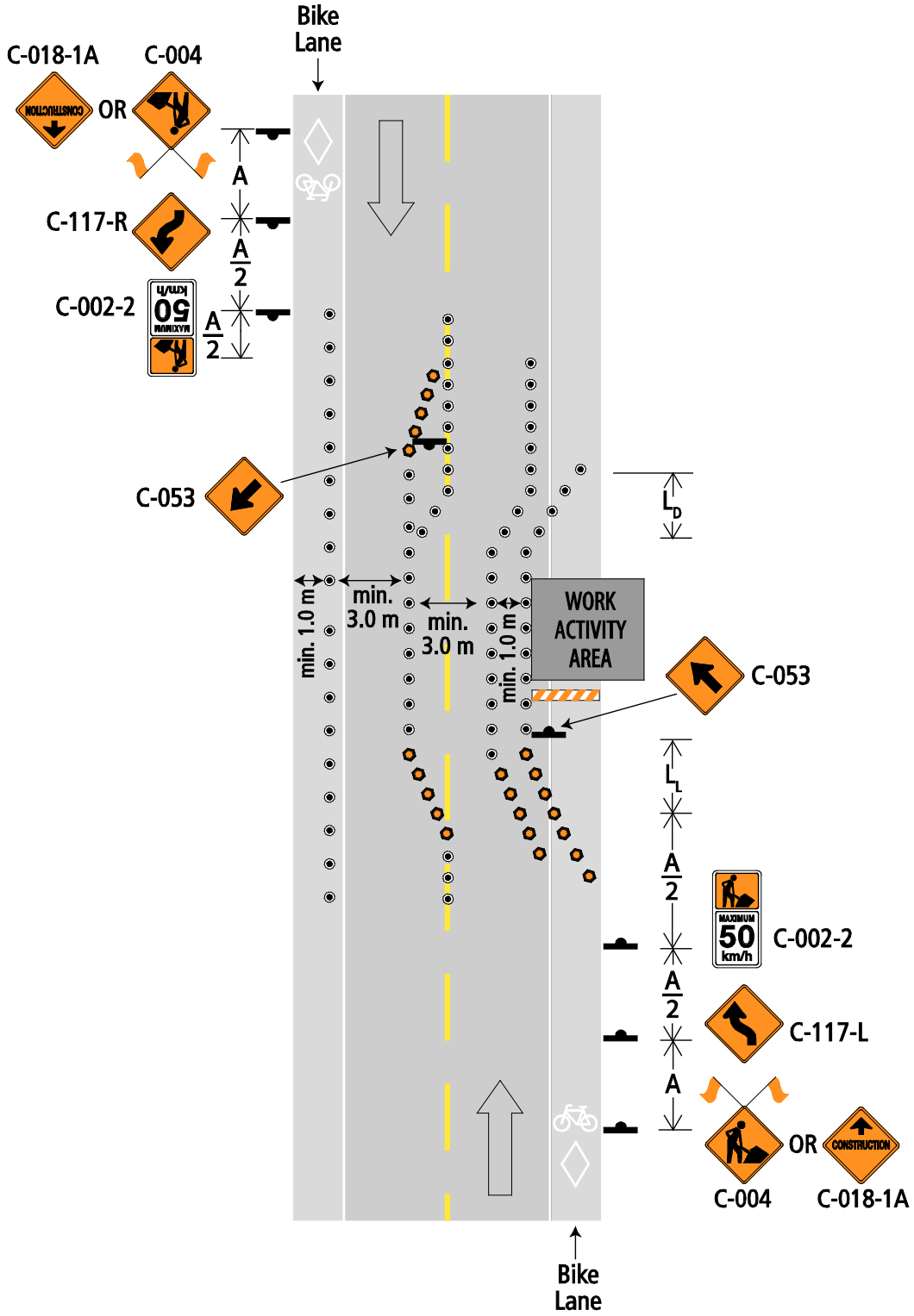
Guidance:

- If standard lane widths cannot be maintained:
  - The vehicle lane may be reduced to a minimum width of 3.0 metres.
  - The bicycle lane may be reduced to a minimum width of 1.0 metre.
- Where there are obstructions adjacent to the open lane, such as barriers, an additional 0.3 metres of bicycle lane width should be provided.
- The bicycle lane should be delineated from the vehicle lane with channelizing devices.
- Where lane widths are reduced, a Construction Speed Zone of 50 km/h or less should be implemented for the section of roadway where lanes are shifted.

Options:

- If the speed limit is ≤ 60 km/h:
  - Tubular markers may be used for leading tapers instead of drums.
  - Cones may be used for protecting the work area.
- For short-duration work when bicycle volumes are low, the bicycle lane may be closed with the following changes to the layout:
  - delineation along the bicycle lane may be removed.
  - a Share the Road W-132-1 sign should be installed upstream of the lane shift.
  - a Bike Lane Closed B-C-002 sign should be installed upstream of the lane shift.

Figure 18.2: Bicycle Lane Shift



18.3 Bicycle Lane Closed – Take the Lane (≤ 50 km/h)

Purpose:

This layout shows the typical setup of a bicycle lane closure in a low-speed urban environment where the regular posted speed is ≤ 50 km/h and a bicycle lane cannot be maintained through the work area because of lane widths or other constraints.

This layout should be used only in areas where the vehicle lane is less than 4.0 metres wide, and there is insufficient lane width for side-by-side bicycle and motor vehicle operation.

It is preferable to allow cyclists to continue cycling when work occurs on a bicycle route, so cyclists are instructed to ride in the centre of the vehicle lane (take the lane) in this layout.

This layout does not apply for shoulder closures.

Standard:

- A Cyclist Right-of-Way "Take the Lane" C-184 sign shall be placed in advance of the taper.
- The speed limit shall be ≤ 50 km/h through the section of roadway where cyclists are taking the lane.

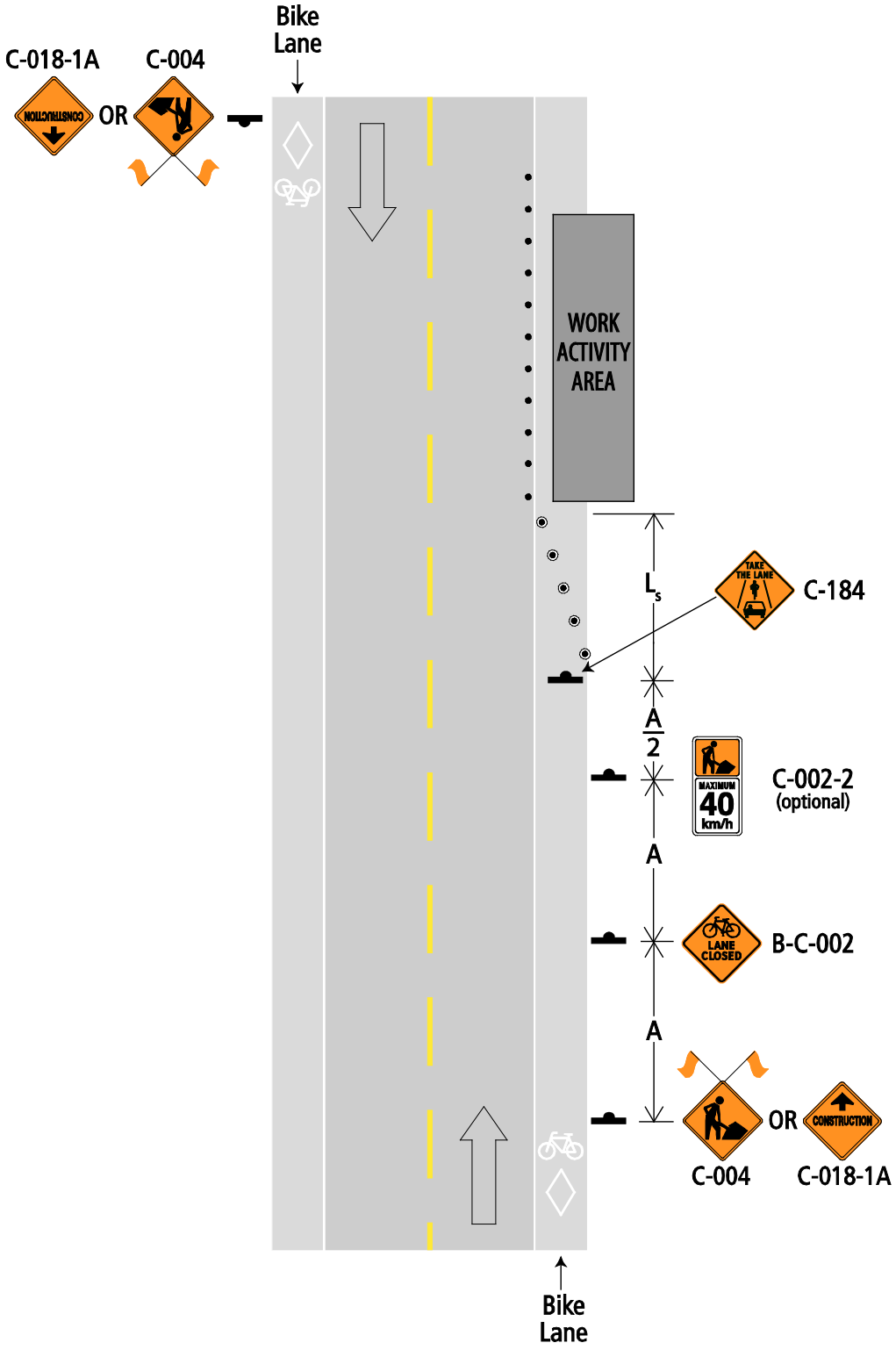
Guidance:

- A reduced Construction Speed Zone should be considered where cyclists are advised to take a vehicle lane.

Options:

- Additional steps may be taken to slow vehicle traffic, such as using a Reduce Speed C-032 sign. See also *Section 2.4: Management of Speed.*
- For closures longer than 150 metres, alternative strategies include:
  - a bicycle lane shift - see *Section 18.2: Bicycle Lane Shift*
  - a bicycle detour - see *Section 18.5: Bicycle Lane Closed – Bicycle Detour*

Figure 18.3: Bicycle Lane Closed – Take the Lane (≤ 50 km/h)



18.4 Bicycle Lane Closed – Share the Road

Purpose:

This layout shows the typical setup for a bicycle lane closure where lane widths allow bicycle traffic to be maintained through the work area by shifting all traffic.

This layout should be used only in areas where lane widths of 4.0 metres or more can be maintained and there is sufficient lane width to have side-by-side bicycle and motor vehicle operation.

This layout does not apply for shoulder closures.

Standard:

- A Road Diversion C-052-L/R sign shall be used to identify the road pattern change.
- A Share the Road W-132-1 sign shall be positioned in advance of the taper.
- A Lane Closure Arrow C-053 sign shall be used where the bicycle and vehicle traffic is being shifted.
- A minimum overall lane width of 4.0 metres shall be maintained.

Guidance:

- Where there are vertical obstructions (e.g., barriers) or drop-offs adjacent to the open lane, an additional 0.3 metres of shy distance should be provided.
- A Construction Speed Zone of 50 km/h or less should be implemented for the section of roadway where bicycles and vehicles are sharing the lane.

Options:

- If there is sufficient lane width, the bicycle space may be delineated from the traffic space with channelizing devices or temporary pavement markings.
- If the speed limit is ≤ 60 km/h:
  - Tubular markers may be used for leading tapers instead of drums.
  - Cones may be used instead of tubular markers.
- A Lane Shift C-117-L/R sign may be used instead of a Road Diversion C-052-L/R sign in the opposite direction.

Figure 18.4: Bicycle Lane Closed – Share the Road

